

vector as follows (where “+” (“−”) indicates a long (short) position respectively):

Security	1	2	...	N
Position	+1	$\hat{\beta}_2$	...	$\hat{\beta}_N$

3. Employ an ECM to provide a mechanism for the spreads deviation from the long-term equilibrium to be corrected. Given the time-series S_A and S_B an ECM can be expressed:

$$\Delta S_t^A = a_1 + \sum_{i=1}^m \beta_{1,1}^i \Delta S_{t-i}^A + \sum_{i=1}^m \beta_{1,2}^i \Delta S_{t-i}^B + \gamma_1 W_{t-1} + \varepsilon_t^1 \quad (13.8)$$

$$\Delta S_t^B = a_2 + \sum_{i=1}^m \beta_{2,1}^i \Delta S_{t-i}^B + \sum_{i=1}^m \beta_{2,2}^i \Delta S_{t-i}^A + \gamma_2 W_{t-1} + \varepsilon_t^2 \quad (13.9)$$

where W is the mispricing term as given above. The lag lengths, as well as the coefficients, can be determined by OLS.

4. As the errors $\varepsilon_t^1, \varepsilon_t^2$ are normally distributed, the vector

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix}$$

defines how the mispricing shall be corrected. The coefficients of this vector correction determine the speed of adjustment. The generalized vector error-correction model (VECM) can thus be represented:

$$\Delta S_t = a + \sum_{i=1}^m \beta_i \Delta S_{t-i} + \sum_{j=1}^q \Gamma W_{t-j} + \varepsilon_t \quad (13.10)$$

where

$$S_t = \begin{pmatrix} s_t^1 \\ \vdots \\ s_t^n \end{pmatrix}, a = \begin{pmatrix} s_1 \\ \vdots \\ a_n \end{pmatrix}, B = \begin{pmatrix} \beta_{1,1} & \cdots & \beta_{1,n} \\ \vdots & \ddots & \vdots \\ \beta_{n,1} & \cdots & \beta_{n,n} \end{pmatrix}$$

$$\Gamma_j = \begin{pmatrix} \gamma^{j1,1} & \cdots & \gamma^{j1,n} \\ \vdots & \ddots & \vdots \\ \gamma^{jn,1} & \cdots & \gamma^{jn,n} \end{pmatrix}, W_t = \begin{pmatrix} w_t^1 \\ \vdots \\ w_t^n \end{pmatrix}, \varepsilon_t = \begin{pmatrix} \varepsilon_t^1 \\ \vdots \\ \varepsilon_t^n \end{pmatrix}$$

which can be estimated via OLS applied to each equation individually.

Once a pair has been identified any deviation from the long-term equilibrium should be corrected through an investment strategy to address the disequilibrium: short the overpriced security and long the underpriced security. The ECM can be used to calculate the length of the mispricing and timings for order execution.